



RTL Design Sherpa

RAPIDS Byte-Granular DMA Engine Byte Characterization Report

September 30, 2026

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Readme

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RAPIDS Byte-Granular DMA: Byte Characterization Report

Version: 0.1

Date: 2026-09-30

Platform: Genesys 2, 256-bit, 8 channels, 100 MHz

Results: rapids_byte_perf_prelim_20260930.json

PRELIMINARY. These numbers were measured on the bitstream that predates the channel-reset fix. They validate the tooling and the report; the final numbers replace them after the rebuild. Cells without a measurement are marked **TBD**.

1. Headline

Largest transfers at 8 channels

Path	Point	Efficiency	MB/s measured	% of 3200 MB/s peak
Sink	byte path, 4096 B x 8 ch	1.000	357	11.1 %
Sink	beat path, 4096 beats x 8 ch	1.000	356	11.1 %
Source	byte path, 4096 B x 8 ch	1.000	352	11.0 %
Source	beat path, 4096 beats	1.000	356	11.1 %

Path	Point	Efficiency	MB/s measured	% of 3200 MB/s peak
x 8 ch				

These rates sit at the harness checker ceiling of $3200 / 9 = 355.6$ MB/s, not at the DUT's limit: the byte-wise CRC checkers take 9 cycles per 32-byte beat (section 3). Read every MB/s in this report against both the 3200 MB/s peak and that ceiling. Efficiency (payload over beats x lanes) does not depend on the checker.

2. Definitions

Definitions

Term	Meaning
Bytes moved	Total payload bytes over all active channels and descriptors; from the exact byte count of the AXIS bus meter.
Beats moved	Accepted beats on the interface named: AXIS beats on the stream side, memory beats on the AXI side.
Efficiency	payload bytes / (beats x BYTE_LANES), BYTE_LANES = 32. 1.0 means every lane of every beat carried payload.
MB/s	Total bytes / measurement window, windows counted in 100 MHz cycles by the harness meters, not wall clock.
Peak	$3200 \text{ MB/s} = 100 \text{ MHz} \times 32 \text{ B}$, per direction. Shown beside every measured MB/s value.
Engaged utilization	$\text{prod} / (\text{prod} + \text{bp} + \text{starv})$ on one interface; the RAPIDS Beats headline metric, used only for the beat-aligned comparison.

3. Beat-aligned rows against the RAPIDS Beats report

These rows use the beat-scaled path (pkt_bytes=None, beats per channel) on the byte DUT, which is how the RAPIDS Beats report measured its matrix: 9-beat bursts, response delay 0,

backpressure off, default seed, bare bus meters. The metric is engaged utilization ($\text{prod} / (\text{prod} + \text{bp} + \text{starv})$), the RAPIDS Beats headline metric. “Unchanged” is checked numerically: every cell is compared to the same cell of `genesys_dw256_obs_C.json` and must agree within 0.5 percentage points, the threshold the RAPIDS Beats report itself applies between builds.

Engaged utilization, byte build / RAPIDS Beats reference, beat-aligned rows

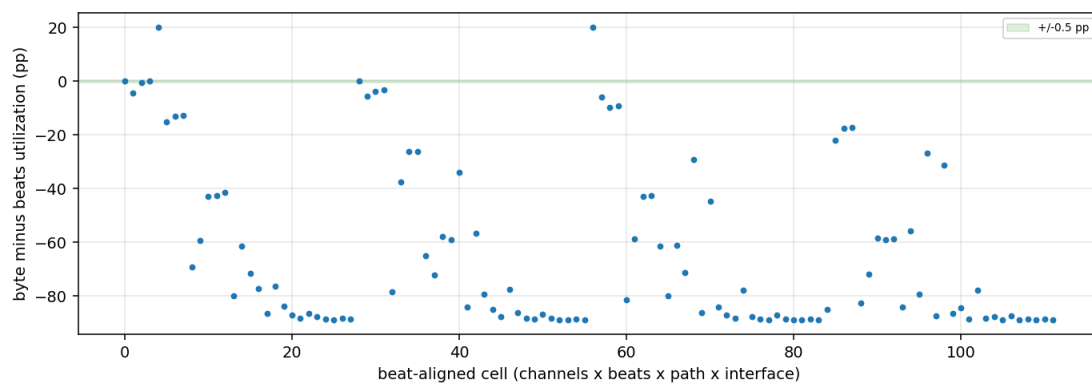
Ch	Beats/ch	Sink AXIS- in % (byte / beats)	Sink AXI4- wr %	Source AXI4-rd %	Source AXIS-out %	Max abs delta (pp)
1	1	50.0 / 50.0	10.0 / 14.3	6.2 / 6.7	6.7 / 6.7	4.29
1	4	100.0 / 80.0	25.0 / 40.0	9.3 / 22.2	9.5 / 22.2	20.00
1	16	25.0 / 94.1	13.6 / 72.7	10.6 / 53.3	10.7 / 53.3	69.12
1	64	57.1 / 98.5	11.6 / 91.4	20.7 / 82.1	10.7 / 82.1	79.79
1	256	22.6 / 99.6	11.2 / 97.7	18.6 / 94.8	11.0 / 94.8	86.47
1	1024	12.7 / 99.9	11.1 / 99.4	12.4 / 98.7	11.1 / 98.7	88.27
1	4096	11.5 / 100.0	11.1 / 99.9	11.4 / 99.7	11.1 / 99.7	88.73
2	1	66.7 / 66.7	16.7 / 22.2	8.0 / 11.8	8.7 / 11.8	5.56
2	4	10.5 / 88.9	19.5 / 57.1	10.1 / 36.4	10.3 / 36.4	78.36
2	16	32.0 / 97.0	12.2 / 84.2	11.7 / 69.6	10.6 / 69.6	72.00
2	64	65.3 / 99.2	11.4 / 95.5	33.6 / 90.1	10.8 / 90.1	84.15
2	256	14.8 / 99.8	11.2 / 98.8	19.8 / 97.3	11.1 / 97.3	87.67
2	1024	11.8 / 100.0	11.1 / 99.7	12.5 / 99.3	11.1 / 99.3	88.58
2	4096	11.3 / 100.0	11.1 / 99.9	11.4 / 99.8	11.1 / 99.8	88.81
4	1	100.0 / 80.0	25.0 / 30.8	9.3 / 19.0	9.8 / 19.0	20.00
4	4	12.9 / 94.1	14.2 / 72.7	10.6 / 53.3	10.7 / 53.3	81.21
4	16	37.2 / 98.5	11.6 / 91.4	21.0 / 82.1	10.8 / 82.1	79.79
4	64	70.3 / 99.6	11.2 / 97.3	50.3 / 94.8	10.8 / 94.8	86.12
4	256	12.6 / 99.7	11.1 / 99.4	20.7 / 98.7	11.1 / 98.7	88.27
4	1024	11.5 / 99.9	11.1 / 99.9	12.6 / 99.7	11.1 / 99.7	88.73

Ch	Beats/ch	Sink AXIS- in % (byte / beats)	Sink AXI4- wr %	Source AXI4-rd %	Source AXIS-out %	Max abs delta (pp)
4	4096	11.2 / 100.0	11.1 / 100.0	11.4 / 99.9	11.1 / 99.9	88.85
8	1	4.0 / 88.9	16.0 / 38.1	10.1 / 27.6	10.4 / 27.6	84.85
8	4	14.5 / 97.0	12.5 / 84.2	11.3 / 69.6	10.6 / 69.6	82.42
8	16	40.5 / 99.2	11.4 / 95.5	34.5 / 90.1	10.8 / 90.1	84.15
8	64	73.1 / 99.8	11.2 / 98.5	66.2 / 97.3	11.0 / 97.3	87.30
8	256	11.8 / 96.1	11.1 / 99.7	21.4 / 99.3	11.1 / 99.3	88.58
8	1024	11.3 / 99.0	11.1 / 99.9	12.6 / 99.8	11.1 / 99.8	88.81
8	4096	11.2 / 99.7	11.1 / 100.0	11.5 / 100.0	11.1 / 100.0	88.87

Verdict: CHANGED. 112 of 112 beat-aligned cells were compared; 108 differ by more than 0.5 pp; the largest difference is -88.87 pp (ch 8, 4096 beats, sink wr). Cells over the threshold: ch1 b1 sink wr -4.29 pp; ch1 b4 sink sin +20.00 pp; ch1 b4 sink wr -15.00 pp; ch1 b4 source rd -12.92 pp; ch1 b4 source sout -12.70 pp; ch1 b16 sink sin -69.12 pp; ch1 b16 sink wr -59.17 pp; ch1 b16 source rd -42.74 pp; ch1 b16 source sout -42.67 pp; ch1 b64 sink sin -41.32 pp; ch1 b64 sink wr -79.79 pp; ch1 b64 source rd -61.34 pp;

What this comparison can and cannot show. The byte build's on-chip checkers (BYTE_CRC=1 on axi4_slave_wr_crc_check and axis4_slave_pattern_check) fold the strobed bytes of each beat into the CRC four bytes per cycle and hold ready low while they do, so a 32-byte beat occupies 9 cycles. That caps the harness at $3200 / 9 = 355.6$ MB/s (11.1 % of the 3200 MB/s peak), which is where the large-transfer rows sit. The RAPIDS Beats reference used the word-wide checkers and is not checker-bound. The large differences above are therefore the checkers backpressuring the DUT, not evidence about the DUT's own throughput, and the "utilization unchanged" criterion cannot be settled on this build. It needs the beat-aligned rows measured with the word-wide checkers (BYTE_CRC=0) on a separate performance bitstream, which also has to keep the 1-beat and 4-beat rows comparable. Until then the beat-aligned rows are **not comparable** to the RAPIDS Beats report.

Figure 3.1: Byte minus beats utilization, beat-aligned cells



utilization delta

Beat-aligned rows: bytes, beats, efficiency, MB/s and share of the 3200 MB/s peak

Ch	Beats /ch	Sink bytes	Sink beats	Sink eff	Sink MB/s	Sink % peak	Src bytes	Src beats	Src eff	Src MB/s	Src % peak
1	1	32	1	1.00 0	320	10.0 %	32	1	1.00 0	200	6.2 %
1	4	128	4	1.00 0	800	25.0 %	128	4	1.00 0	298	9.3 %
1	16	512	16	1.00 0	430	13.4 %	512	16	1.00 0	339	10.6 %
1	64	2048	64	1.00 0	372	11.6 %	1056	33	1.00 0	342	10.7 %
1	256	8192	256	1.00 0	359	11.2 %	4864	152	1.00 0	353	11.0 %
1	1024	32768	1024	1.00 0	357	11.1 %	29440	920	1.00 0	355	11.1 %
1	4096	131072	4096	1.00 0	356	11.1 %	127744	3992	1.00 0	355	11.1 %
2	1	64	2	1.00 0	533	16.7 %	64	2	1.00 0	256	8.0 %
2	4	256	8	1.00 0	337	10.5 %	256	8	1.00 0	324	10.1 %
2	16	1024	32	1.00	389	12.2	928	29	1.00	340	10.6

Ch	Beats /ch	Sink bytes	Sink beats	Sink eff	Sink MB/s	Sink % peak	Src bytes	Src beats	Src eff	Src MB/s	Src % peak
				0		%			0		%
2	64	4096	128	1.00	363	11.4	1312	41	1.00	344	10.8
				0		%			0		%
2	256	1638	512	1.00	357	11.2	9152	286	1.00	354	11.1
		4		0		%			0		%
2	1024	6553	2048	1.00	356	11.1	5830	1822	1.00	355	11.1
		6		0		%	4		0		%
2	4096	2621	8192	1.00	356	11.1	2549	7966	1.00	356	11.1
		44		0		%	12		0		%
4	1	128	4	1.00	800	25.0	128	4	1.00	298	9.3
				0		%			0		%
4	4	512	16	1.00	413	12.9	512	16	1.00	339	10.6
				0		%			0		%
4	16	2048	64	1.00	372	11.6	1056	33	1.00	346	10.8
				0		%			0		%
4	64	8192	256	1.00	358	11.2	1760	55	1.00	346	10.8
				0		%			0		%
4	256	3276	1024	1.00	357	11.1	1753	548	1.00	355	11.1
		8		0		%	6		0		%
4	1024	1310	4096	1.00	356	11.1	1158	3620	1.00	355	11.1
		72		0		%	40		0		%
4	4096	5242	1638	1.00	356	11.1	5090	1590	1.00	356	11.1
		88	4	0		%	56	8	0		%
8	1	256	8	1.00	129	4.0	256	8	1.00	324	10.1
				0		%			0		%
8	4	1024	32	1.00	397	12.4	960	30	1.00	338	10.6
				0		%			0		%
8	16	4096	128	1.00	363	11.4	1280	40	1.00	345	10.8
				0		%			0		%
8	64	1638	512	1.00	357	11.1	2720	85	1.00	352	11.0
		4		0		%			0		%

Ch	Beats /ch	Sink bytes	Sink beats	Sink eff	Sink MB/s	Sink	Src bytes	Src beats	Src eff	Src MB/s	Src % peak
						% peak					
8	256	6553	2048	1.00	356	11.1	3392	1060	1.00	355	11.1
		6		0		%	0		0		%
8	1024	2621	8192	1.00	356	11.1	2305	7204	1.00	356	11.1
		44		0		%	28		0		%
8	4096	1048	3276	1.00	356	11.1	1016	3178	1.00	356	11.1
		576	8	0		%	960	0	0		%

Every beat-aligned row moves whole beats, so efficiency is 1.000 by construction; measured range over 56 rows: 1.000 to 1.000.

4. Transfer size (offset 0)

One descriptor per channel, payload in bytes, start offset 0. Bytes moved and beats are totals over the active channels; efficiency is payload bytes over beats times 32 bytes per beat, on the AXIS side and on the memory side. MB/s is total bytes over the measurement window (the longer of the stream and memory windows) and is always shown with its share of the 3200 MB/s peak (100 MHz x 32 B).

4.1 Sink path

Size sweep, sink path, offset 0

Payl oad B	Offset	Ch	Bytes move d	AXIS beats	Mem beats	Eff (AXIS)	Eff (mem)	MB/s	% of peak	Result
1	0	1	1	1	1	0.031	0.031	10.0	0.31 %	PASS
1	0	2	2	2	2	0.031	0.031	16.7	0.52 %	PASS
1	0	4	4	4	4	0.031	0.031	25.0	0.78 %	PASS
1	0	8	8	8	8	0.031	0.031	4.0	0.13 %	PASS
7	0	1	7	1	1	0.219	0.219	70.0	2.19 %	PASS
7	0	2	14	2	2	0.219	0.219	116.7	3.65 %	PASS

Payl oad B	Offset	Ch	Bytes move d	AXIS beats	Mem beats	Eff (AXIS)	Eff (mem)	MB/s	% of peak	Result
7	0	4	28	4	4	0.219	0.219	175.0	5.47 %	PASS
7	0	8	56	8	8	0.219	0.219	28.1	0.88 %	PASS
32	0	1	32	1	1	1.000	1.000	320.0	10.00 %	PASS
32	0	2	64	2	2	1.000	1.000	533.3	16.67 %	PASS
32	0	4	128	4	4	1.000	1.000	800.0	25.00 %	PASS
32	0	8	256	8	8	1.000	1.000	128.6	4.02 %	PASS
33	0	1	33	2	2	0.516	0.516	275.0	8.59 %	PASS
33	0	2	66	4	4	0.516	0.516	471.4	14.73 %	PASS
33	0	4	132	8	8	0.516	0.516	113.8	3.56 %	PASS
33	0	8	264	16	16	0.516	0.516	129.4	4.04 %	PASS
64	0	1	64	2	2	1.000	1.000	533.3	16.67 %	PASS
64	0	2	128	4	4	1.000	1.000	914.3	28.57 %	PASS
64	0	4	256	8	8	1.000	1.000	220.7	6.90 %	PASS
64	0	8	512	16	16	1.000	1.000	251.0	7.84 %	PASS
77	0	1	77	3	3	0.802	0.802	550.0	17.19 %	PASS
77	0	2	154	6	6	0.802	0.802	208.1	6.50 %	PASS

Payload B	Offset	Ch	Bytes moved	AXIS beats	Mem beats	Eff (AXIS)	Eff (mem)	MB/s	% of peak	Result
77	0	4	308	12	12	0.802	0.802	256.7	8.02 %	PASS
77	0	8	616	24	24	0.802	0.802	290.6	9.08 %	PASS
203	0	1	203	7	7	0.906	0.906	369.1	11.53 %	PASS
203	0	2	406	14	14	0.906	0.906	431.9	13.50 %	PASS
203	0	4	812	28	28	0.906	0.906	386.7	12.08 %	PASS
203	0	8	1624	56	56	0.906	0.906	367.4	11.48 %	PASS
1024	0	1	1024	32	32	1.000	1.000	389.4	12.17 %	PASS
1024	0	2	2048	64	64	1.000	1.000	371.7	11.62 %	PASS
1024	0	4	4096	128	128	1.000	1.000	363.4	11.36 %	PASS
1024	0	8	8192	256	256	1.000	1.000	359.5	11.23 %	PASS
4035	0	1	4035	127	127	0.993	0.993	360.9	11.28 %	PASS
4035	0	2	8070	254	254	0.993	0.993	358.0	11.19 %	PASS
4035	0	4	16140	508	508	0.993	0.993	356.1	11.13 %	PASS
4035	0	8	32280	1016	1016	0.993	0.993	355.6	11.11 %	PASS
4096	0	1	4096	128	128	1.000	1.000	363.4	11.36 %	PASS
4096	0	2	8192	256	256	1.000	1.000	359.5	11.23 %	PASS

Payl oad B	Offset	Ch	Bytes move d	AXIS beats	Mem beats	Eff (AXIS)	Eff (mem)	MB/s	% of peak	Result
409 6	0	4	1638 4	512	512	1.000	1.000	357.5	11.17 %	PASS
409 6	0	8	3276 8	1024	1024	1.000	1.000	356.5	11.14 %	PASS

4.2 Source path

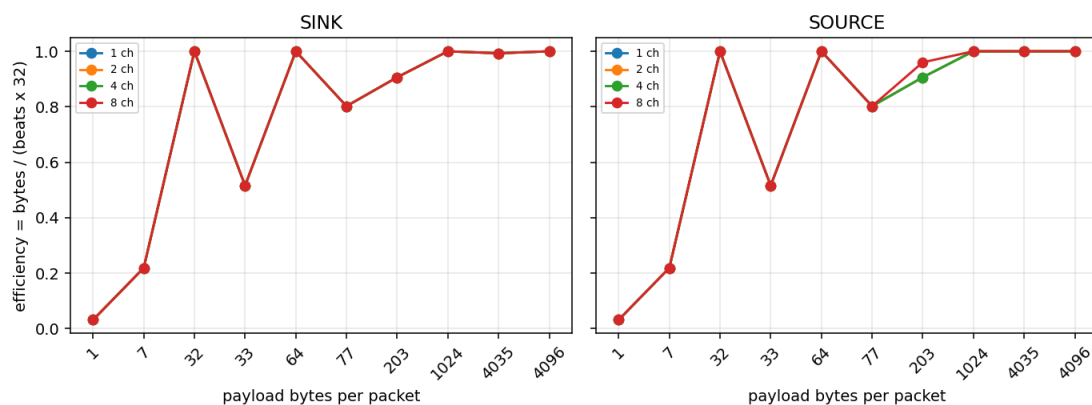
Size sweep, source path, offset 0

Payl oad B	Offset	Ch	Bytes move d	AXIS beats	Mem beats	Eff (AXIS)	Eff (mem)	MB/s	% of peak	Result
1	0	1	1	1	1	0.031	0.031	6.2	0.20 %	PASS
1	0	2	2	2	2	0.031	0.031	11.1	0.35 %	PASS
1	0	4	4	4	4	0.031	0.031	18.2	0.57 %	PASS
1	0	8	8	8	8	0.031	0.031	26.7	0.83 %	PASS
7	0	1	7	1	1	0.219	0.219	43.8	1.37 %	PASS
7	0	2	14	2	2	0.219	0.219	73.7	2.30 %	PASS
7	0	4	28	4	4	0.219	0.219	112.0	3.50 %	PASS
7	0	8	56	8	8	0.219	0.219	151.4	4.73 %	PASS
32	0	1	32	1	1	1.000	1.000	200.0	6.25 %	PASS
32	0	2	64	2	2	1.000	1.000	256.0	8.00 %	PASS
32	0	4	128	4	4	1.000	1.000	297.7	9.30 %	PASS

Payload B	Offset	Ch	Bytes moved	AXIS beats	Mem beats	Eff (AXIS)	Eff (mem)	MB/s	% of peak	Result
32	0	8	256	8	8	1.000	1.000	324.1	10.13 %	PASS
33	0	1	33	2	2	0.516	0.516	132.0	4.12 %	PASS
33	0	2	66	4	4	0.516	0.516	183.3	5.73 %	PASS
33	0	4	132	8	8	0.516	0.516	227.6	7.11 %	PASS
33	0	8	264	16	16	0.516	0.516	258.8	8.09 %	PASS
64	0	1	64	2	2	1.000	1.000	256.0	8.00 %	PASS
64	0	2	128	4	4	1.000	1.000	297.7	9.30 %	PASS
64	0	4	256	8	8	1.000	1.000	324.1	10.13 %	PASS
64	0	8	512	16	16	1.000	1.000	339.1	10.60 %	PASS
77	0	1	77	3	3	0.802	0.802	226.5	7.08 %	PASS
77	0	2	154	6	6	0.802	0.802	270.2	8.44 %	PASS
77	0	4	308	12	12	0.802	0.802	299.0	9.34 %	PASS
77	0	8	616	24	24	0.802	0.802	315.9	9.87 %	PASS
203	0	1	203	7	7	0.906	0.906	290.0	9.06 %	PASS
203	0	2	406	14	14	0.906	0.906	317.2	9.91 %	PASS
203	0	4	812	28	28	0.906	0.906	332.8	10.40 %	PASS

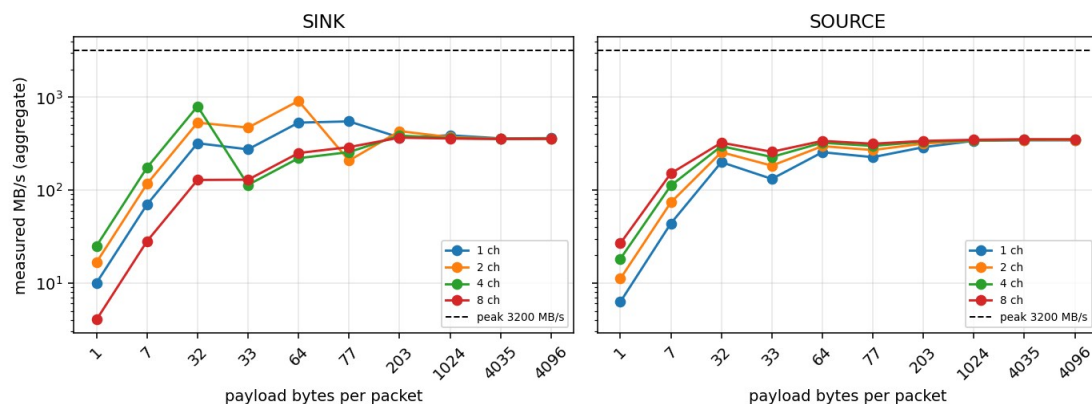
Payload B	Offset	Ch	Bytes moved	AXIS beats	Mem beats	Eff (AXIS)	Eff (mem)	MB/s	% of peak	Result
203	0	8	1014	33	56	0.960	0.566	339.1	10.60 %	PASS
102 4	0	1	928	29	32	1.000	0.906	339.9	10.62 %	PASS
102 4	0	2	1056	33	64	1.000	0.516	341.7	10.68 %	PASS
102 4	0	4	1280	40	128	1.000	0.312	343.2	10.72 %	PASS
102 4	0	8	1760	55	256	1.000	0.215	348.5	10.89 %	PASS
403 5	0	1	1280	40	127	1.000	0.315	344.1	10.75 %	PASS
403 5	0	2	1760	55	254	1.000	0.217	347.1	10.85 %	PASS
403 5	0	4	2656	83	508	1.000	0.163	349.0	10.91 %	PASS
403 5	0	8	4512	141	1016	1.000	0.139	353.3	11.04 %	PASS
409 6	0	1	1280	40	128	1.000	0.312	344.1	10.75 %	PASS
409 6	0	2	1760	55	256	1.000	0.215	347.1	10.85 %	PASS
409 6	0	4	2688	84	512	1.000	0.164	352.3	11.01 %	PASS
409 6	0	8	4512	141	1024	1.000	0.138	352.0	11.00 %	PASS

Figure 4.1: Efficiency against payload size



efficiency vs size

Figure 4.2: Measured MB/s against payload size, peak as the dashed line



MB/s vs size

Observations from this data:

- Sink, 8 channels, 4096 B: 357 MB/s = 11.1 % of the 3200 MB/s peak at efficiency 1.000.
- Sink, 8 channels: 1 B payload reaches 4.0 MB/s, 89x less than 4096 B; the window is dominated by fixed per-packet latency (199 cycles for one beat per channel).
- Source, 8 channels, 4096 B: 352 MB/s = 11.0 % of the 3200 MB/s peak at efficiency 1.000.
- Source, 8 channels: 1 B payload reaches 26.7 MB/s, 13x less than 4096 B; the window is dominated by fixed per-packet latency (30 cycles for one beat per channel).

5. Start offset

A descriptor may start mid-beat. The offset moves payload into an extra memory beat when offset + payload crosses a beat boundary, lowering memory-side efficiency while the AXIS side, which packs from lane 0, is unchanged. Offset 0 rows repeat the size sweep.

5.1 Sink path

Offset sweep, sink path

Payl oad B	Offset	Ch	Bytes move d	AXIS beats	Mem beats	Eff (AXIS)	Eff (mem)	MB/s	% of peak	Result
33	0	1	33	2	2	0.516	0.516	275.0	8.59 %	PASS
33	1	1	33	2	2	0.516	0.516	275.0	8.59 %	PASS
33	5	1	33	2	2	0.516	0.516	275.0	8.59 %	PASS
33	31	1	33	2	2	0.516	0.516	275.0	8.59 %	PASS
203	0	1	203	7	7	0.906	0.906	369.1	11.53 %	PASS
203	1	1	203	7	7	0.906	0.906	TBD	TBD	FAIL
203	5	1	203	7	7	0.906	0.906	369.1	11.53 %	PASS
203	31	1	203	7	8	0.906	0.793	225.6	7.05 %	PASS
1024	0	1	1024	32	32	1.000	1.000	389.4	12.17 %	PASS
1024	1	1	1024	32	33	1.000	0.970	317.0	9.91 %	PASS
1024	5	1	1024	32	33	1.000	0.970	318.0	9.94 %	PASS
1024	31	1	1024	32	33	1.000	0.970	324.1	10.13 %	PASS
4035	0	1	4035	127	127	0.993	0.993	360.9	11.28 %	PASS

Payload B	Offset	Ch	Bytes moved	AXIS beats	Mem beats	Eff (AXIS)	Eff (mem)	MB/s	% of peak	Result
4035	1	1	4035	127	127	0.993	0.993	360.9	11.28 %	PASS
4035	5	1	4035	127	127	0.993	0.993	361.2	11.29 %	PASS
4035	31	1	4035	127	128	0.993	0.985	344.6	10.77 %	PASS
33	0	8	264	16	16	0.516	0.516	129.4	4.04 %	PASS
33	1	8	264	16	16	0.516	0.516	129.4	4.04 %	PASS
33	5	8	264	16	16	0.516	0.516	129.4	4.04 %	PASS
33	31	8	264	16	16	0.516	0.516	129.4	4.04 %	PASS
203	0	8	1624	56	56	0.906	0.906	367.4	11.48 %	PASS
203	1	8	1624	56	56	0.906	0.906	TBD	TBD	FAIL
203	5	8	1624	56	56	0.906	0.906	368.3	11.51 %	PASS
203	31	8	1624	56	64	0.906	0.793	249.8	7.81 %	PASS
1024	0	8	8192	256	256	1.000	1.000	359.5	11.23 %	PASS
1024	1	8	8192	256	264	1.000	0.970	328.6	10.27 %	PASS
1024	5	8	8192	256	264	1.000	0.970	328.7	10.27 %	PASS
1024	31	8	8192	256	264	1.000	0.970	329.5	10.30 %	PASS
4035	0	8	32280	1016	1016	0.993	0.993	355.6	11.11 %	PASS

Payl oad B	Offset	Ch	Bytes move d	AXIS beats	Mem beats	Eff (AXIS)	Eff (mem)	MB/s	% of peak	Result
403 5	1	8	3228 0	1016	1016	0.993	0.993	355.6	11.11 %	PASS
403 5	5	8	3228 0	1016	1016	0.993	0.993	355.7	11.12 %	PASS
403 5	31	8	3228 0	1016	1024	0.993	0.985	347.4	10.86 %	PASS

5.2 Source path

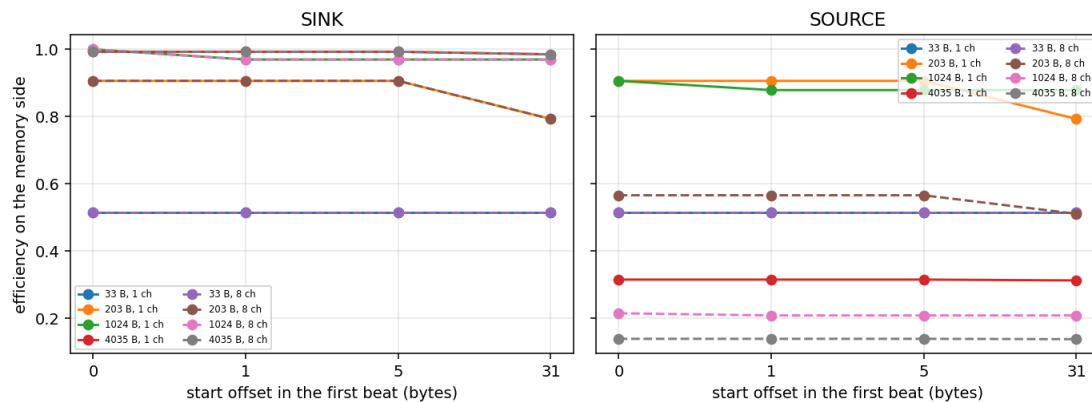
Offset sweep, source path

Payl oad B	Offset	Ch	Bytes move d	AXIS beats	Mem beats	Eff (AXIS)	Eff (mem)	MB/s	% of peak	Result
33	0	1	33	2	2	0.516	0.516	132.0	4.12 %	PASS
33	1	1	33	2	2	0.516	0.516	126.9	3.97 %	PASS
33	5	1	33	2	2	0.516	0.516	126.9	3.97 %	PASS
33	31	1	33	2	2	0.516	0.516	126.9	3.97 %	PASS
203	0	1	203	7	7	0.906	0.906	290.0	9.06 %	PASS
203	1	1	203	7	7	0.906	0.906	285.9	8.93 %	PASS
203	5	1	203	7	7	0.906	0.906	285.9	8.93 %	PASS
203	31	1	203	7	8	0.906	0.793	285.9	8.93 %	PASS
102 4	0	1	928	29	32	1.000	0.906	339.9	10.62 %	PASS
102 4	1	1	928	29	33	1.000	0.879	339.9	10.62 %	PASS

Payload B	Offset	Ch	Bytes moved	AXIS beats	Mem beats	Eff (AXIS)	Eff (mem)	MB/s	% of peak	Result
1024	5	1	928	29	33	1.000	0.879	339.9	10.62 %	PASS
1024	31	1	928	29	33	1.000	0.879	339.9	10.62 %	PASS
4035	0	1	1280	40	127	1.000	0.315	344.1	10.75 %	PASS
4035	1	1	1280	40	127	1.000	0.315	344.1	10.75 %	PASS
4035	5	1	1280	40	127	1.000	0.315	344.1	10.75 %	PASS
4035	31	1	1280	40	128	1.000	0.312	344.1	10.75 %	PASS
33	0	8	264	16	16	0.516	0.516	258.8	8.09 %	PASS
33	1	8	264	16	16	0.516	0.516	256.3	8.01 %	PASS
33	5	8	264	16	16	0.516	0.516	256.3	8.01 %	PASS
33	31	8	264	16	16	0.516	0.516	256.3	8.01 %	PASS
203	0	8	1014	33	56	0.960	0.566	339.1	10.60 %	PASS
203	1	8	1014	33	56	0.960	0.566	339.1	10.60 %	PASS
203	5	8	1014	33	56	0.960	0.566	339.1	10.60 %	PASS
203	31	8	1046	34	64	0.961	0.511	344.1	10.75 %	PASS
1024	0	8	1760	55	256	1.000	0.215	348.5	10.89 %	PASS
1024	1	8	1760	55	264	1.000	0.208	345.1	10.78 %	PASS

Payload B	Offset	Ch	Bytes moved	AXIS beats	Mem beats	Eff (AXIS)	Eff (mem)	MB/s	% of peak	Result
1024	5	8	1760	55	264	1.000	0.208	345.1	10.78 %	PASS
1024	31	8	1760	55	264	1.000	0.208	345.1	10.78 %	PASS
4035	0	8	4512	141	1016	1.000	0.139	353.3	11.04 %	PASS
4035	1	8	4512	141	1016	1.000	0.139	353.3	11.04 %	PASS
4035	5	8	4512	141	1016	1.000	0.139	353.3	11.04 %	PASS
4035	31	8	4512	141	1024	1.000	0.138	352.0	11.00 %	PASS

Figure 5.1: Memory-side efficiency against start offset



efficiency vs offset

Offsets that cost an extra memory beat (sink, 1 channel): 203 B at offset 31: 7 -> 8 memory beats; 1024 B at offset 1: 32 -> 33 memory beats; 1024 B at offset 5: 32 -> 33 memory beats; 1024 B at offset 31: 32 -> 33 memory beats; 4035 B at offset 31: 127 -> 128 memory beats.

6. Descriptor chains

Several descriptors per channel back to back (4 channels), offset 0. Efficiency counts payload over all beats of the chain.

Descriptor chains, 4 channels

Payload B	Descs	Path	Bytes moved	AXIS beats	Eff (AXIS)	MB/s	% of peak	Result
1	16	sink	64	64	0.031	5.4	0.17 %	PASS
1	16	source	64	64	0.031	37.2	1.16 %	PASS
64	4	sink	1024	32	1.000	223.6	6.99 %	PASS
64	4	source	1024	32	1.000	347.1	10.85 %	PASS
64	16	sink	4096	128	1.000	221.4	6.92 %	PASS
64	16	source	3872	121	1.000	352.6	11.02 %	PASS
1024	4	sink	16384	512	1.000	341.4	10.67 %	PASS
1024	4	source	2656	83	1.000	350.9	10.96 %	PASS
1024	16	sink	65536	2048	1.000	336.3	10.51 %	PASS
1024	16	source	50432	1576	1.000	355.4	11.11 %	PASS

7. Source backpressure (integrity only)

The source backpressure is paced by the host toggling the checker's ready over UART, so the cycle counts measure the host, not the DUT (rapids TASK-085). These rows are integrity checks: bytes and beats must still match and the golden CRCs must agree. Their MB/s is not reported.

Source with backpressure, integrity rows

Ch	Payload B	Offset	Expected bytes	Expected AXIS beats	MB/s	Result
1	33	0	33	2	n/a (host-paced)	PASS
1	33	5	33	2	n/a (host-paced)	PASS
1	1024	0	1024	32	n/a (host-paced)	PASS

Ch	Payload B	Offset	Expected bytes	Expected AXIS beats	MB/s	Result
1	1024	5	1024	32	n/a (host-paced)	PASS
4	33	0	132	8	n/a (host-paced)	PASS
4	33	5	132	8	n/a (host-paced)	PASS
4	1024	0	4096	128	n/a (host-paced)	PASS
4	1024	5	4096	128	n/a (host-paced)	PASS

8. Failures, caveats and coverage

Failed points as recorded, never dropped

Point	Path	After earlier failure	Errors (first two)
offset_ch1_p203_o1_d1	sink	-	ch0: SINK GOLDEN MISMATCH (DUT DATA BUG) wr=0x850D0C5F golden=0x67E11839
offset_ch8_p203_o1_d1	sink	offset_ch1_p203_o1_d1	ch0: SINK GOLDEN MISMATCH (DUT DATA BUG) wr=0x850D0C5F golden=0x67E11839; ch1: SINK GOLDEN MISMATCH (DUT DATA BUG) wr=0xBD687A55 golden=0x5F846E33

A failure is recorded with its errors and is never retried away. Points after the first failure carry the id of that failure (`after_failure`): the sticky sink packet-length flag and the AXI response error flags clear only on `aresetn`, and `CHANNEL_RESET` does not reach them, so one failed point can poison the ones after it. A failure that is repeatable at its own coordinates, with no earlier failure in the file, is a genuine result.

Coverage: 105 of 105 planned points of profile standard are in this file (103 passed, 2 failed).

Standing limitations of the design, not of this measurement:

- The sink `s_axis_tready` is one signal qualified by TID, so a beat for a channel whose packet record has not arrived blocks every channel behind it on the stream (head-of-line blocking, inherent and documented).
- `TYPE=EXT` descriptors stay beat-aligned by design and are not part of the byte sweeps.
- Each interface has its own measurement window; the `sin` window runs from the first to the last stream beat (rapids ISSUE-001), so windows differ per interface and MB/s here uses the longer of the stream and memory windows.

9. Provenance and reproduction

Provenance

Item	Value
Results file	<code>rapids_byte_perf_prelim_20260930.json</code>
Timestamp	2026-09-30T09:40:04
Profile	standard
Status	PRELIMINARY
Bitstream	<code>rapids_byte.bit</code> sha256 2de9c03d369b6014
Design	256-bit, 8 channels, 4096 B SRAM per channel, 100 MHz, peak 3200 MB/s per direction
Beats reference	<code>genesys_dw256_obs_C.json</code>

No device readback is recorded in this results file: it was measured before the campaign began reading `CSR_ID`, `BUILD` and the configure-time sentinel at the start and end of each session. The bitstream identity above is the file hash the run was started from, not a readback, so a reprogram by another user during the run would not have been detected. The final run records the readback.

One command runs the campaign and regenerates this report:

```
cd projects/fpga-systems/Genesys2/rapids/flows-rapids
./byte_perf.sh --profile standard          # PRELIMINARY: writes
*_prelim_*.json
./byte_perf.sh --profile full --final      # final numbers, after the
channel-reset fix is on the board
```